TRISTAN MEYNIER

meyniertristan10@gmail.com | https://github.com/tmeynier | tristanmeynier.com

Passionate ML and Software Engineer – 5 years of experience
Strong technical skills. Fluent in English, French and German.

EDUCATION

- **ETHZ:** MSc in *Data Science* (Sep 2019 – Sep 2021) - **GPA:** 5.34 / 6.00
- **Carnegie Mellon University:** *Computer Science* (Aug 2018 – Jun 2019) - **GPA:** 4.00 / 4.00, Dean's List Award
- **EPFL:** BSc in *Computer Science* (Sep 2016 – Jul 2018) - **GPA:** 5.48 / 6.00

TECHNICAL SKILLS

- **Programming Languages:** Python (PyPI packaging, Claude code development, GitLab CI/CD, Conventional Commits, Dockerization, Code Formatting with Black, Documentation with Sphinx), Scala and Java
- **Frameworks & Libraries:** PyTorch, Pandas, Scikit-learn
- **Tools & Platforms:** Docker, Linux, AWS, Azure (Synapse & ML Studio), Airflow
- **Data:** PostgreSQL, MongoDB, Databricks, Dataiku, PySpark, RunPod
- **Application & API Development:** Django, Node.js, React, Streamlit, Dash, Flask, FastAPI

EXPERIENCE

Expedia Group - ML Scientist II – Geneva

Feb 2026 – Present (1 year)

- Developed statistical and machine learning models in Python to optimize Expedia pricing for hotels and flights, with the objective of maximizing revenue and bookings.
- Models and methods: constrained non-linear optimization, tree-based models (CatBoost, LightGBM), GLMs, and neural networks.
- Use Codex and Claude to interact with Databricks clusters and Trino SQL engines, enabling large-scale analyses on datasets exceeding 1B rows.
- Design and analyze A/B tests using confidence intervals, p-values, and statistical power analysis.
- Use relevant econometric and business signals, including price elasticity, propensity scores, conversion rates, and booking probabilities.
- Work within a team of 10 ML scientists, contributing across the full model lifecycle from R&D and experimentation to production deployment on Airflow.
- Present results to stakeholders in accessible terms, with a focus on business impact, assumptions, and model limitations.

Hepha AI - Startup Founder & ML Engineer – Zürich

June 2023 – Present (3.5 years)

- Research project and startup exploring state-of-the-art Vision-Language-Action policies for controlling robotic arms and autonomous mobile robots in warehouse-like environments.
- Read and analyzed research papers and evaluated state-of-the-art robot policies on custom synthetic and robotic manipulation tasks.
- Developed and trained custom AI policies in PyTorch on self-collected datasets, using cloud GPUs on RunPod, dataset and model versioning on Hugging Face, code versioning on GitLab, and Weights & Biases for experiment tracking.
- Documented the project on my website (tristanmeynier.com/hepha) and in research reports: tristanmeynier.com/what-makes-a-generalist-robot-policy and tristanmeynier.com/what-robot-data-should-we-collect-next.
- Deployed policies through a Django–React application and automated model deployment using Docker and CI/CD pipelines.
- Showcased the project through a Django Progressive Web App, Blender animations, and custom 3D-printed robotic components.

Syngenta - Senior ML Engineer – Basel
Oct 2024 – Present

- Conducted applied research to evaluate the quality and business value of AI models (reviewed research papers, formulated and tested hypotheses).
- Developed and published Python packages exposing APIs for serving AI models.
- Conceptualized, sketched, and presented the IT infrastructure (databases, backend, frontend) to support AI models in production (real-time model inference).
- Worked in an Agile team of 7 ML engineers and developers, using Jira for collaboration and GitLab CI/CD pipelines for automated builds, testing, and pull request (PR) workflows.
- Maintained code quality and integration through unit testing, code formatting, and Sphinx documentation.
- Presented project PoC (Streamlit app) in regular department meetings.

KPMG - (Senior) Data Scientist – Zürich
October 2021 to October 2024

- Designed a machine learning time-series forecasting pipeline for pricing optimization at Hewlett-Packard using LightGBM on Dataiku with PySpark. This 1.5-year project encompassed data analysis, model testing, visualization, and production deployment.
- Implemented an end-to-end ML pipeline for an automated financial risk prediction tool on Python with Azure Cloud (Synapse & ML Studio) using NLP with pre-trained HuggingFace Sentence Transformers.

IBM Research – Zürich
Master Thesis in Machine Learning *Mars 2021 - October 2021*

- Thesis outline: reconstruct the 3D genome structure from biological data using Deep Learning.
- A significant part of the work included interviews with SMEs and reading papers.
- Improved the mean squared error prediction of previous works by 70% using transfer learning on PyTorch.

Genistat AG - Zürich
Part-time Machine Learning Engineer *September 2020 - Mars 2021*

- Genistat is a start-up applying data science on media data.

Virtual Traffic Lights - Pittsburgh
Part-time (and full-time) Software Developer *August 2018 - September 2019*

- Virtual Traffic Lights is a start-up using AI for better car traffic management.

LEADERSHIP AND AWARDS

- **Certifications:** Azure A-Z 900, Azure DP-100, Azure AI Engineer Associate
- Selected among the 7 top students of the CS department at EPFL for an exchange year at Carnegie Mellon.

PUBLICATIONS

- **(Preprint)** *Improving Chemical Impurity Prediction via Bootstrapped Reaction Ensembles*.
- **LMRL:** *Modeling the 3D Chromatin Structure from Hi-C Data with Transfer Learning*.
- **ICLR:** *Recovering Barabási-Albert Parameters of Graphs through Disentanglement*.

ABOUT ME

- Creative and enterprising; eager to learn new skills. Chess enthusiast. Running and hiking lover!